HARDWARE IP DISPUTE VERIFICATION

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CERTIFICATE

This is to certify that the work embodied in this report entitled **"HARDWARE IP DISPUTE VERIFICATION"** has been satisfactorily completed by **SHUBHAM SAURAV (24U020005)** and **MANISH KUMAR (24U020039)** It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering**, **Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2024-28.

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DECLARATION

We hereby declare that the following major project synopsis entitled “HARDWARE IP DISPUTE VERIFICATION” presented in the is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in** Computer Science and Engineering. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Rahul Chaurasia**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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AREA OF WORK

This project focuses on Hardware security, chip design and cryptography. Nowadays companies around the world work together to manufacture computer chips which increases the risk of chip design being copied or used without permission.

Traditional security method uses secret key for verifying the ownership, however these key may need to revealed during disputes and it can create the privacy and security risks.

To solve this problem our project uses a public- private key security system to verify ownership without revealing confidential information. It embeds unique digital watermarks into the chip design during High Level Synthesis using register allocation and Conflict Interference Graph (CIG) constraints.

The main advantage is about protecting the chips and rights of the original owner by preventing them from sharing the direct secret to the public at the time of dispute along with almost no additional hardware overhead, no extra processing delay or no extra power consumption for this newly introduced design.

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ABSTRACT

Hardware piracy is a major problem in semiconductor industry, as the chip design can be stolen by other rival companies and they can sell them by showing their brand to the customer. There are many way to keep the original chip design safe, one of them is watermarking i.e, embedding the secrets directly into the design of chip, so that other company could not be able to reverse the design and open up the secret, but the existed watermarking technique can either make the design/circuit slower or force the owner to reveal their private secret key at the time of dispute.

This project presents a practical solution using biometric patterns to protect the Intellectual Property and the design of the hardware chip. The unique points from the biometric patterns are used and turned into rules that controls how registers are assigns in the circuit. This results in no overhead to the circuits i.e, an invisible watermark is embedded in the circuit with zero costs to the circuit size and speed. At the time of verification of suspected chip, the design is checked using asymmetric method, public rules let an verifier to check a suspected chip on a website without revealing the secret information.

To solve the security issue of sharing secrets to verifier at the time of verification, the system uses an asymmetric verification method. A public key is created so that anyone can check a chip is genuine or pirated without seeing the original signature and private rules. An end-to-end web systems was developed using FastAPI to let uses register designs, test suspect chip by uploading register mappings and verify the origin of the suspected chips. Tests shows that the system can correctly find copied or stolen designs, with a chance of making a wrong match is very less.

INTRODUCTION

The main chip used in the modern electronic devices like smartphones, televisions, computers etc, are rarely made and designed from scratch as building a complex chip takes too much time and money. So company uses pre-built digital building blocks known as Hardware Intellectual Property cores.

While reusing the IP cores makes the production of devices easy, but it may lead to a security problem i.e, it may causes Hardware piracy. The designs of the chip are shared through software-like code (e.g-RTL files), so it can happen that the dishonest companies or untrusted manufacturer can easily copy the design and place it inside their own chip and will sell them by their own brand name without paying the license fees. This illegal copying of design costs too much to the semiconductor industries every year.

The problem with the traditional solution:

Traditional techniques uses the method of watermarking embedded inside the digital circuit. However the older watermarking methods have two major issues:

1. They waste chip resources: The standard methods adds extra logic gates or dummy circuits just to hold the watermark. These extra overhead makes the chip larger, slower and more power consuming.

2. The “Inspection Paradox” : Most older system uses symmetric verification, this means that the secret key used to hide inside the chip design through watermarking is the exact same secret key used later to prove the ownership of the suspected chip. Therefore this results in leakage of security information.

If the owner takes a suspect chip to court or the verifier to prove it was stolen, than they are forced to show their private key. The moment that secret is exposed, than anyone can use it to remove watermark or can copy design. Therefore in short by traditional method you cannot prove the ownership without revealing your secret information about the original chip design.

Our Solution : Biometric secret and Asymmetric verification

To solve both problem at once, this project introduces a practical zero overhead protection system based on two main ideas:

1. Using the human biometric as the Signature: Instead of using a random password or text string, we are extracting the secure signature from human biometric and this can be either fingerprint, face or retina scan. We turns the unique point of biometric patterns to the unique digital signature that cannot be guessed or recreated by the attacker.

2. Zero Overhead Embedding: Every digital circuits has internal variables that hold numbers temporarily during calculations. During the chip design stage i.e, during High Level Synthesis phase, our system uses the signature to decide which variables share the same register. The circuit experiences zero penalty in chip size , speed or power consumption, and this is because we only organize how existing registers are shared without adding any extra gates or wires.

3. Asymmetric Verification : We have solved the problem of secret leaking in old technique by splitting the verification in two halves:

- Private Embedding: The owner uses the secret biometric pattern to build the circuit.
- Public checking: Using cryptographic one-way mathematical commitments, IP owner generates a public verification profile linked to the private signature constraints.

Anyone such as independent auditor or court can upload a suspect chip register layout to our public website and checks if the rules match and will get verdict as genuine or pirated.

Key Contribution of this work:

- Natural Biometric Watermark:
- 100% Zero Overhead:
- Safe Public Verification:
- Working web platform:

Github Repository link:
https://github.com/shubhamsaurav15107-art/minor_project

LITERATURE REVIEW

This chapter gives brief information about the 23 research papers on zero-knowledge proofs(ZKPs) and related cryptographic techniques. The review has the main purpose to place this project in context. The goal of this project is to verify the pirated version of original hardware chip by verifying the vendors signature constraint to prove the ownership of the suspected design.

The primitive existed techniques need the signature and the full constraint list, and revealing them exposes their secret information to the rival companies. Therefore our project combines signed, salted commitment with selective opening, plus a Pedersen-commitment-variant, so that a vendor can prove the true origin of their hardware without disclosing the secrets.

Section 2 gives the hardware-security background. Section 3 summarises each of the 23 papers with its source, a short description and the main conclusion, followed by a line on what it means for this project. Section 4 draws the papers together and stated the gap the project addresses.

1. Background: hardware IP watermarking and zero-knowledge detection

Systems-on-chip combine IP cores licensed from many vendors, so an IP owner has little control once a core has been delivered. A watermark embeds a signature in the core without changing its function, so that ownership can be shown later. It is a passive measure: it can prove that an IP was used, but it cannot prevent infringement .

The base paper for this project [24] follows register- allocation route and derives the signature from a biometric. Branching and bifurcation points are located on a binarised retinal images by convolution, their Mahattan distance values are concatenated int a binary signature, and each symbol selects a pair of storage variables that must not share a register: an odd-odd pair for 1, for an even-even pair for 0 and a Zero integer pair for the binary point. The authors report no increase in register count. They compare the scheme with earlier digital-signature,fingerprint,facial and palmprint approaches.

2. Review of the 23 Papers:

[1] Analysing Security and Efficiency in a Quantum- Resistant Decentralised E-Voting System using blockchain,Homomorphic Encryption and Zero-knowledge Proofs.
A. Kumar, V.M. Shrimal and R.Rubbina, RMKMATE 2025, Chennai,India.
Electronic voting is quick and transparent but open to manipulation and data breaches, and ECC and RSA based schemes would fall to quantum computers. The author combine blockchain, homomorphic encryption and ZKPs with post-quantum techniques for secure, verifiable and private voting.
Conclusion: this system gives strong benefits, with legal and computational challenges still open and future works covers hybrid cryptographic models, AI- based fraud detection, quantum-safe consenus and regulatory acceptance.
Relevance: ZKPs give verifiabilty without disclosure of any information and the same here in this paper , as it applies to the signature and Pedersen commitment used in this Project, whereas hash- based commitment are less exposed.

[2] Effiecient Smart Contract for Privacy- Preserving Authentication in Blockchain Using Zero-Knowledge Proof.
D. Naidu et al., OCIT 2023, Raipur, India, pp. 969-974.
Phishing, replay and password attacks have grown with advancement in digital payments. The paper uses ZKP based authentication on blockchain so users prove they hold a credential without revealing it, keeps payment record unchanged and can be tracked. It also demonstrates a smart contract for fund flow between vendors.
Conclusion: The mechanisms gives secure authentication and access control in private blockchain and protection from phishing, man-in-the middle and password- guessing attacks.
Relevance: The project needs a permanent records from its registry that proves a commitment was made before any dispute happened.

[3] Securing Records in Healthcare lo T using Zero- knowledge Proof Based Blockchain.
S. Mishra and R.K. Dwivedi, ICICAT 2025, Gorakhpur, India, pp. 1-5.
Traditional health records system store data in one central place and use password, so if that system fails or get hacked, the data may be at risk.
The framework uses smart contracts to keep track of activities, IPFS to store large files, and Circom and SnarksJS to prove the eligibility of user.
Conclusion: It results in decentralised storage, ZKPs and loT keeps records private and tamper- proof while reducing blockchain cost.
Relevance: Circom with SnarksJS are the tools to add zero knowledge proofs to this project.

[4] Zero Knowledge Programming Cerifications.
P. Baptista et al.., ICDLT 2025, Pune, India, pp, 1-5.
This platform extracts data from a GitHub accounts, such as bytes written per languages and generates a zero knowledge proof that any party trusting the platform can verify without seeing the repositories. Proving can might take little time but the verification can runs on laptop and phones.
Conclusion: the prototype met its goals (checking the programming skill of a applicants) despite slow proving and limits in the GitHub data.
Relevance: This is similar to proving something using private data without revealing itself. This paper trusts the platform that collects the data, and is similar to our project where it trust the person who creates the register map for the suspected chip.

[5] Zero Knowledge One Time Digital Signature Scheme.
N. Datta, IEEE Int. COnf. on Computational Intelligence and Computing Research, Coimbatore, India, 2012, pp. 1-4.
This is a ZKP based algorithm that lets a reciever confirms a messages's integrity without the sender sending the signature directly for applications where integrity matters more than secrecy. The same documents gives a different signature on every transmission, so the scheme acts as a one time signature.
Conclusion: It becomes harder to tamper the message without transmitting and makes it more secure by using the document's hash as the key.
Relevance: Our projecct also use a similar idea of confirming integrity without handling over the signature. Here the evidence is a hardware watermark instead of a document dignature, but the purpose is same.

[6] Privacy Preserving Authentication Using Zero Knowledge Proofs.
A. Abraham et al., IEEE CICN 2025, Goa, India, pp. 1003-1008.
Passwords and One time passwords can be stolen or exposed through hacking and phishing. The paper suggest a password free login system where the users have to prove that they know the secret without storing or sharing the secret itself.
Conclusion: a Schnorr ZKP based scheme was implemented and shown to justify completeness, soundness and zero-knowledge and using new values on each login prevent replay and leakage of password.
Relevance: Schnorr identification is a common method for proving knowledge of a secret key. This project uses fresh random values for each login to prevent attackers for reusing the old data.

[7] LocalVault: Zero knowledge Fingerprint Authentication without Biometric storage:
Q. Akash et al., ICCCI 2026, Okayama, Japan, pp. 238-243.
Storing fingerprint can be risky because fingerprints cannot be changed if they are once stolen. LocalVault uses a fuzzy extractor to create a stable secret from fingerprint reading. It stores only helper data and commitment, not the fingerprint itself.
Conclusion: This sytem had successfully rebuilt the fingerprint in 92% of attempts and took less than one second to respond.
Relevance: This is the closest example because it uses a biometric-based secret, and a commitment, and zero knowledge proof without showing the original biometric data to verifier. However LocalVault is used to verify person's identity, while this project proves that a hardware design contains a mark created from biometric data.

[8] A paradigm shift in Digital Trust: A Cybersecurity Analysis of Zero- knowledge Proofs.
V.Bobade et al., CMSS 2026, Shah Alam, Malaysia, pp. 1573-1576.
This study explains how ZKP improves security, privacy and data accuracy in blockchain,identity management and loT system. It also discusses challenges and the complexity of implemeting ZKPs.
Conclusion: ZKP help protect privacy while allowing information to be verified in areas such as identity management , blockchain, loT, finance, and e-governance. However, challenges remain, including high computing cost, legal issues, and difficult in making different systems work together.
Relevance: Its obstacles also apply to hardware IP, where this review found no established practice for proving a watermark in zero knowledge, therefore usage is limited.

[9] Zero Knowledge Proofs for Privacy- Preserving Agricultural Yield Verification : A Blockchain - Based Incentive System for Sustainable Farming Methods.
P.Singhal and N.Joshi, IC3 2025, Noida, India, pp 1-6.
An Ethereum-based reward system where farmers prove they have met their crop yield targets without disclosing private farming data.
Conclusion: the ZKPIncentive Mechanishm works as expected. In the future it could considers more factor than just crop yield, and uses cheaper zero knowledge proof methods and also allows users to participate in decision making through decentralized governance.
Relevance: It shows a threshold- style statement proven without underlying data, which has the same shape as proving that opened rules hold without revealing the rest.

[10] Research on the Recovery Strategy of Federated Leraning Model Based on Zero - Knowledge proof.

X. Wang, AAICV 2025, London, UK, pp. 194 -198.
Traditional privacy techniques are not enough for federated learning. The paper uses zero knowledge, Euclidean distance and cossing similarity, to check whether machine learning models have been tampered with by malicious users, without revealing private data.
Conclusion: the method improves privacy and protects the system against the attcaks, however the study does not show how efficient this method is.
Relevance: it proves in zero knowledge that private values satisfy public constraints. A circuits for this project has the same form, checking that the registers of each hidden variable pair differ.

[11] if-ZKP: Intel FPGA- Based Acceleration of Zero Knowledge Proofs.
S.A. Butt et all., IEEE FCCM 2024, Orlando,USA, p. 212.
zk-SNARK proving is slow mainly because of multi- scalar multiplication therefore the authors build an FPGA architecture with Intel oneAPI that exploits MSM parallelism and optimised modular arithmetic.
Conclusion: The FPGA based design performs better than traditional CPU and can match a GPU. It is especially effective for large mathematical operations, offering faster processing and better energy efficiency.
Relevance: This paper uses FPGAs to speed up the creation of zero-knowledge proofs, while this project uses zero-knowledge proofs to verify a hardware design.

[12] Comparative study of Diverse Zero-Knowledge Argument Systems.
J. Kuriakose et al., SAPIENCE 2016, Ernakulam, India, pp. 284-293.
This paper compares interactive, non- interactive, resettable and concurrent zero- knowledge systems, including their behaviour against a malicious prover and malicious verifier.
Conclusion: A perfect zero-knowledge system may not be possible for every type of problem under certain theoretical conditions. Also using ZKPs can increase the systems processing time and computational costs.
Relevance: It frames two design questions. First, the prototype requires interaction, because the server issues a random challenge and could become non- interactive with the Fiat- Shamir heuristic. Second, the system must protect against dishonest verifiers as repeated challenges against one design would reveal more of the signature, so the number of checks must be limited.

[13] zkSHIELD - A Web-Based Cross-Chain Zero Knowledge Asset Proof.
V.K. Polisetti et al.,ICCCNT 2024 , Kamand,India,pp.1-5.
Users may have their digital assets spread across blockchains and wallets such as Ethereum , Bitcoin and Polygon. The web application displays these assets, links secondary wallets and tracks past transfer requests, keeping for each transfer the content identifier of a zk-SNARK proof stored on IPFS so that it can be accessed later.
Conclusion: Financial transparency can be offered while keeping sensitive information private, with support for selective disclosure, data integrity , user control and scalability.
Relevance : This is another web application that uses zero-knowledge proofs to share only the necessary information while keeping the private data hidden. It also stores proof for future checking, similar to how this project keeps a registry of signed commitment for later verification.

[14] Digital Identity Management Based on Blockchain and Zero Knowledge Proof Under the PIPL.
M.Chen et al., CSCWD 2026, Fuzhou, Chaina, pp.340-345.
Traditional identity management does not satisfy China's personal information Protection Law. The scheme generates proofs off-chain, verifies them on-chain and uses dynamic authorisation so users stay in control of their data.
Conclusion: This system performs more better and efficient than existing solutions. However it has some limitations like- ZKP overhead on constrained devices and weak cross-chain compatibility.
Relevance: A retinal image is sensitive personal data, so data-protection law applies to any system that stores it. in this project the server never receives the retinal image and sees signature bits.

[15] ZCube: A Zero-Trust, Zero-knowledge, and Zero-Memory Platform for Privacy and yet Secured Access.
V. Quach et al., IEEE TPS-ISA 2024,Washington, DC, USA, pp. 166-175.
The traditonal System stores information to build trust are becoming less reliable. ZCube uses nested ZKPs with a blockchin-based trustless setup to verify entities in real time without storing past data and each request is verified independently.
Conclusion: A continuously running decentralized, memoryless platform with bidirectional verification, practical for microservices.
Relevance: Each verification session on this project's website is independent, and the server holds no vendor secret between sessions. the response times suggest ZK-style checks can fit interactive web use.

[16] An Alternative Methodology for Authentication and Confidentiality Based on Zero Knowledge Protocols using Diffie-Hellman Key Exchange.
P.L.S Kumari and A. Damodaram, ICIT 2014, Bhubaneswer,India,pp. 368-373.
A zero-knowledge protocol concept is mixed with Diffie-Hellman key exchange so a user proves knowledge of a password without sending it through mutual identification and a random session key.
Conclusion:The system focuses on verifying users, performing calculations efficiently, using less memory, and preventing attackers from making privacy dispute.
Relevance: The group-based proofs with the elliptic-curve cryptography which supports Pedersen commitments and digital signatures. The impersonation point matches the vendor signature's role here and only the key holder can register a design or open its rules.

[17] Secure Authentication Using Zero knowledge Proof.
A. Pathak et al., ASIANCON 2021, Pune, india, pp. 1-8.
The system uses Secure Remote Password (SRP) protocol which allows users to log in without sending their actual password over the network. It also uses AES encryption o protect data and improves security compared with traditional username and password login systems.
Conclusion: SRP with AES gives stronger security and privacy and is simple enough to integrate into web applications.
Relevance: It shows zero knowledge based login integrated into a web application, as this project's website does with signed requests.

[18] Secure Digital Service payments Using Zero Knowledge Proof in Distributed Network.
M. Harikrishan and K.V. Lakshmy, ICACCS 2019, Coimbatore, India, pp. 307-312.
Blockchains allow fair exchange without a trusted third party, but confidentiality blocks usage of enterprise. Zcash's zk-SNARK needs a trusted setup, while zk-STARK does not and resists quantum attacks. The paper proposes a system using two indistinguishable hash functions from the SHA-3 finalists together with zk-STARK .
Conclusion: ZKPs gives confidentiality on a blockchain, STARKs avoid the trusted setup and are post-quantum secure, though SNARKs are faster and interactive Oracle Proofs need improvement for scale.
Relevance: This helps the project choose between two types of zero-knowledge proofs: SNARKs, which usually require a trusted setup, and STARKSs which used high based methods and do not require a trusted setup. STARKs may fit well with the project's existing hash-based commitments.

[19] UAV-Assisted Zero Knowledge Model Proof for Generative AI: A MUltiagent Deep Reinforcement Learning Approach.
M. Hao et al., IEEE Internet of Tings J., vol. 12, no. 10, pp. 13441-13454, May 2025.
The users of generative AI cannot confirm model quality and also providers will not reveal models details so this paper proposes a UAV-based non-interactive ZKP architecture that verifies key GAN operations without exposing parameters, with an age of Verification metric and a multiagent PPO algorithm.
Conclusion: The parameters stay protected and quality is checked efficiently and future works looks at other ZKP methods and may look into more deeper data-security issues.
Relevance: Its main benefit is that the proof can be checked without a live verifier. Similarly using the Fiat-Shamir method, this project could turns its interactive challenges into a non interactive proof.

[20] Defending Location Privacy Using Zero- Knowledge Proof Concept in Location Base Services.
P. Jagwani and S. Kuashik, IEEE MDM 2012, Bengaluru, India, pp. 368-371.
The paper links the authentication techniques of an earlier middleware architecture to zero-knowledge proofs, reducing dependence on trusted third party.
Conclusion: The scheme suits pull- based services while push -based services remain open also it still needs trusted middleware, so future work aims at third party free ZKP authentication.
Relevance: The same trust issue appears here in this project, the project's server is a semi trusted party because it records timestamps registrations and runs checks. Storing these timestamp on a public blockchain would reduce the need to trust the server and provide a more reliable record of when registration occurs.

[21] Zero Knowledge Protocol in a Radio over Fiber Network:
G.L. de Oliveira et al., SBMO/IEEE MTT-S IMOC 2009, Belem, Brazil, pp. 834-837.
A radio-over-fiber netwrok in which several verifier users run a zero knowledge protocol with one prover, with an experiment on photonic microwave frequency doubling.
Conclusion: the setup can support a quantum zero-knowledge service, with two different frequency carriers with the two wireless users can take part.
Relevance: This paper concerns a physical-layer network and therefore it is kept for completeness and can be dropped if the chapter needs to be shorter.

[22] Privacy Protection Revolution: Zero-Knowledge Proof:
S.Liu, ICDACAI 2022, Zakopane,Poland, pp. 394-397.
A review of the history of verification and secrecy and of ZKP theory and its time cost, followed by zk-SNARK with linear-algebra and polynomial implementations and applications in Zcash, Filecoin.
Conclusion: ZKP and zk-SNARK opens the way to decentralised systems on solid mathematics, but implementations bugs and risks are there.
Relevance: It supplies the theory behind the planned SNARK extension, and its warning about implementation bugs supports this project's emphasis on tests and independent review.

[23] Efficient Zero-Knowledge Identification Scheme with Secret Key Exchange.
A.M. Alliam et al., 46th Midwest Symp. on Circuits and Systems, Cairo, Egypt, 2003, pp. 516-519.
Challenge- response identification can leak partial information, and a hostile verifier can pick challenges to extract a secret. The authors has presented a Fiat-Shamir-like scheme on the basis of elliptic curve discrete logarithm problem along with secret key exchange and a mutual identification extension.
Conclusion: This work gives equal security with much smaller keys as compared to RSA and the scheme is faster than Fiat-Shamir as it uses addition instead of exponentiation.
Relevance: It belongs to the discrete logarithm with Pedersen commitments and Schnorr-style proofs.

4. Synthesis

4.1 What the project takes from each theme

- **Zero-knowledge identification and authentication [5],[6],[16],[17],[23].** The three formal properties (completeness, soundness, zero knowledge) as evaluation criteria and fresh randomness in every session is there. The discrete-logarithm constructions behind Pedersen commitments and signatures are built in our project and integrated into web applications.
- **Biometric, identity and access control[7], [14], [15], [20].** A commitment-based use of a biometric secret without holding it in the clear [7] along with data protection duties for biometric data [14] and a smooth stateless per session checks [15]. The cost of depending on a trusted middleman is taken from this[20].
- **Blockchain applications [1], [2], [3], [13], [18].** The immutable, timestamped registries [2]; the Circom and SnarksJS toolchain [3] and also selective disclosure with stored proof records [13]. The SNARK versus STARK choice and trusted setup had been analysed[18] and about the quantum concerns [1].
- **Proving a property of hidden data or models [4], [9], [10], [19].** The patterns of proving a statement about private data without revealing it, and the split between proving cost and verification cost.
- **Foundations, surveys and performance [8], [11], [12], [21], [22].** The limits of perfect zero knowledge and its cost [12]. The SNARK theory and bug risk [22]; the MSM bottleneck and FPGA acceleration [11]; a general list of obstacles [8]. Paper [21] has little bearing on the project.

4.2 The gap this project addresses

Across all this 23 papers we have seen that ZKPs are applied to authentication, identity, blockchains, machine learning and network protocols. None of theme had addresses hardware IP, and no paper proves the presence of a watermark in a design. The hardware watermarking treats verifications as something the owner performs with full knowledge of the secret. As far as this review found, the combination of a biometric-derived constraint watermark in register allocation with a way to show its presence without full disclosure has not been studied.

4.3 Limits that carry over into the project

- The prototype performs selective disclosure, not a formal zero-knowledge proof. Opened rules are visible, and each reveals the signature bit at its position. Full zero knowledge needs a SNARK or STARK, and [4], [10] and [11] suggest proving cost will mater for hundreds of constraints.
- Evidence strength depends on how many rules are opened. The chance that an unmarked design with c registers passes k rules by luck is Pc=(1-1/c)^k, the coincidence formula. A few rules give weak evidence.
- A server-countersigned timestamp makes registration order checkable only if the server is trusted [20]; anchoring on a public ledger [2] would remove that assumption.
- Elliptic-curve signatures and Pedersen commitments are not quanatum-resistant [1]; hash based commitments are less exposed.
- A suspect chip's register map must be expressed in the vendor's variable numbering and whoever builds it must be trusted by both sides, as with the data extraction in [4].

5. Summary

The 23 papers shows that zero-knowledge techniques are mature enough for authentication, identity and blockchain use, with known costs in proving time and implementation risk. The hardware IP literature has solid methods for embedding and testing marks but treats verification as full disclosure. This project joins the two ideas on a small scale: it keeps the signature and its constraint list sealed, opens only the blinding factor at the time of verification, and names a SNARK- based proof as the route to a complete zero-knowledge guarantee.

PROBLEM DEFINITION AND OBJECTIVES

In the modern electronics industry, chip designer engineers are rarely creating an entire integrated circuit from very basic. They simply use the already built digital blocks call Semiconductor Intellectual Property -IP Cores. These IP cores are delivered as digital code or netlists. So there may be chances that the dishonest third party or hackers can easily steal the design and pirate it and may sell it by changing the brand name without a proper license about them.
There are already watermarking schemes for protecting the chip design but it has two drawbacks:
1. **Hardware Resource Overhead:** We need to add extra things in circuit for proving at the time of dispute but it leads to slow down the chip's performance and may consumes extra battery power.
2. **The Inspection Paradox (Symmetric Key Flaw):** The most of the older system uses symmetric verification, this means that the secret key used to hide inside the chip design through watermarking is the exact same secret key used later to prove the ownership of the suspected chip. Therefore this results in leakage of security information.

Objectives:

The main objectives of this project are:

- **Unique biometric source:** This paper can uses the three biometric methods for generating the security constraints. It is fingerprint, facial data, and retinal data, and among them retinal data is the most secure method of security.
- **Asymmetric Verification:** This paper has used a method in which using cryptographic commitment and digital signatures, verifiers can easily check the authentic chip without revealing the vendor's private biometric template.
- **Interactive Full-Stack Web Platform:** The goal of this project is to make a public interactive client-server application that manages registration of the designs, can have features of verification of suspected chips, and also any trusted verifier can sees the authenticity of a particular chip.

PROPOSED METHODOLOGY AND WORK DESCRIPTION

Overview of the Architecture:

The system connects three independent attribute: Vendor, Server and the verifier.

- **Vendor:** These are the owner of the chip. They generates an cryptographic key pair and converts the digital signatures into register allocation rules, and later they seals each rule inside a cryptographic commitment after that they publishes only the commitment and public key on the server.
- **Server:** It act as a neutral trusted party just like the court that delivers decision by making themselves as a neutral body. The server host the public design registry in database with a secure owner public key, design netlist hash, digital signatures and pedersen commitment. At the time of verification the verifier checks the cryptographic proofs by independently verifying the RSA signature with the registered public key and recomputing the commitment using the revealed blinding factor.
- **Verifier (Auditor/Buyer):** These are the people or auditor who wants to verify whether a suspected physical chip contains the proprietary IP core. Verifiers need to submits the suspect chip's register map to the web portal and reads the results without needing any login or private keys.

The features and unique points are extracted from biometric scan and converted into binary string. The binary coordinates form a long string contains three symbols '0', '1', and '.' (representing the decimal point). After this the rules are decided for the extracted constraints i.e, which variables are to be stored in which register. The vendor maps each character of the signature into a constraint of the form "Variable V_i and Variable V_j must not share the same register" using the designated pool. Like vendor may decide even even pool for the string '0', odd-odd pool to string '1' and '.' to Zero integer pool.

The Asymmetric Verification Protocol:

To solve the problem of inspection paradox, the system uses asymmetric verification by following these steps:

1. Rules Sealing (Commitment Phase): The vendor commits the secret watermark bitstream and selects a random blinding factor r using pedersen commitment:

$$\text{Commitment: } C = h*G + r*H$$

where G,H are elliptical curve point and h is the hashing of bitstream.

The vendor compute the SHA 256 hash of the circuit design netlist hash and digitally signs using RSA-2048 private key. After that vendor transmits only the commitment C, digital signature, the netlist hash and the corresponding public key. The vendors private key and the blinding factor are kept confidential at this phase.

2. Dispute Verification Phase (Reveal Phase) : In this phase the auditor or court can verifies the ownership of a suspected chip IP core without exposing the vendors private key. The working of this phase is stated below:

1. Extraction and Evidence Opening:
    - The watermark is extracted from the suspected netlist's register table.
    - The vendors reveal the blinding factor r and along with this the already published public key is used at this verification.

2. Cryptographic checks:

- Pedersen commitment check: The auditor computes the commitment and verifiers both the original and suspected commitment. This proves that the watermark was committed before the dispute and not fabricated later.
- Public key Signature check: The auditor verifies the digital signature using the public key: $\text{Verify}_{\text{PuK}}(\text{Hash}||\text{ Bitstream, Signature})$
  This authenticates the vendor's authorship and ensures the netlist integrity.

3. Verdict:

If both the commitment and the signature match, than the system confirms the IP violation. The server provides a good platform for indisputable proof of the ownership without compromising the vendor's private credentials.

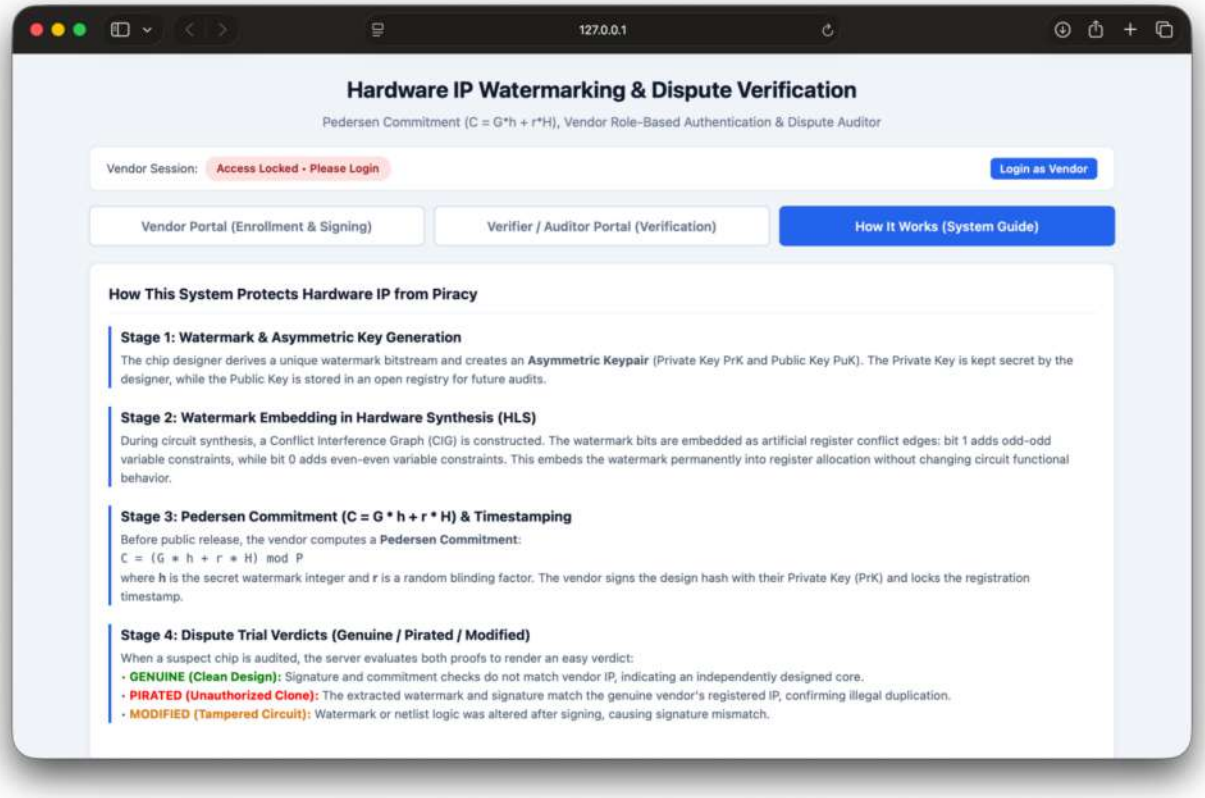


Figure 1. How web platform works

PROPOSED ALGORITHMS

The proposed algorithms in securing the hardware IP is such that it validate ownership using asymmetric public key cryptography and cryptographic commitments and is stated below:

Stage 1: Key & Watermark generation:

The IP owner generate the secret digital constraint using the biometric pattern and derives the secret pattern.

1. Secret bitstream: A secret bit stream is derived from the vendors signature template (b_1,b_2,b_3,…….b_n).

2. Asymmetric key Pair: The vendor generates an asymmetric key pair (Private Key, Public Key). The private key is kept secret but the public key is registered to the website database. Also we have used RSA-2048.

3. Pedersen Commitment: This is a algorithm by which owner can protect their secret bits and also it locks the watermark as evidence without sharing secrets so at the time of dispute owner can prove the ownership of the chip.

   Commitment: C = h*G + r*H

4. Cryptographic hash function: SHA-256 has been used for creating a fixed length of the RTL netlist and payloads before digital singing.

After this register allocation graph is prepared for the chip IP and the secrets gets embedded inside the chip during High Level Synthesis phase. CIG construction using Data Flow Graph is built and artificial edges are inserted in the register design.

Stage 2: Dispute Resolution:

On getting any suspected chip design in the market the suspected design goes under the different phase before coming on any conclusion about the chip. These verification steps are stated below:

1. Structural Analysis: The suspected design goes under the experimental phase through which the tester recovers the bitstream through the register allocation table of the chip.
2. Cryptographic Verification: The auditor evaluates the design using the public credentials.
   - Verify the digital signatures using the public key of vendor

     $Verify_{PuK}$(Hash|| Bitstream, Signature)
   - Verify the commitment :

     C = h*G + r*H
3. Verdict: Infringement is confirmed if the signature and commitment match and structural and functional properties of design exceeds the certain limits.

PROPOSED FLOWCHART

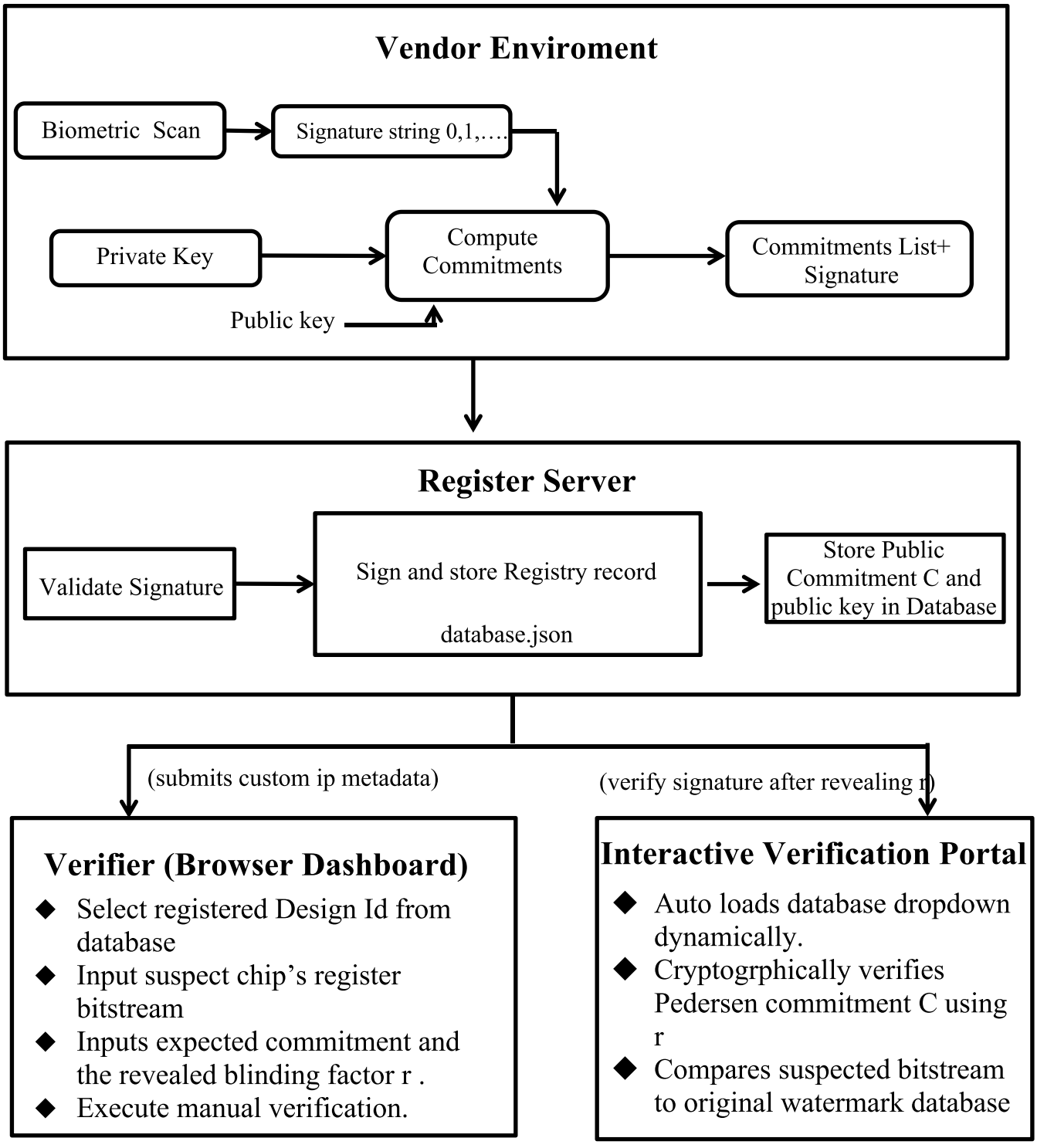


Figure 2. Flowchart

IMPLEMENTATION (TECHNOLOGY USED)

This system is implemented as a full stack architecture across client, server and user interface layers:

Table 1: Tools used in this project

Component	Technology	Specification	Role
Backend	Python	3.9+	Server logic and cryptographic evaluation
Web sever	Flask	>=2.0.0	It helps in API and routing
Asymmetric cryptography	RSA	2048 bit key length	It generates private and public key for verification
Hash Engine	SHA-256	256 bit digest	It creates netlist integrity and payload hashing
Data Storage	Local json	File structure	It helps to load demo IP automatically from *databse.json*
Frontend UI	HTML5/CSS3/Js	-	It designs an interactive web platform for vendor registration and verification of design.

This project has used python as a backbone for making backend of the website such that this backbone controls the mathematical part of this project such as commitment and asymmetric design phase. Flask API is used to make the web platform design user interactive. SHA-256 had been used for converting the bitstream to hash. We have used a file named *database.json* to store the information about IP for demo purpose on our machine locally and all this are connected to our frontend which is prepared using HTML, CSS and javascript.

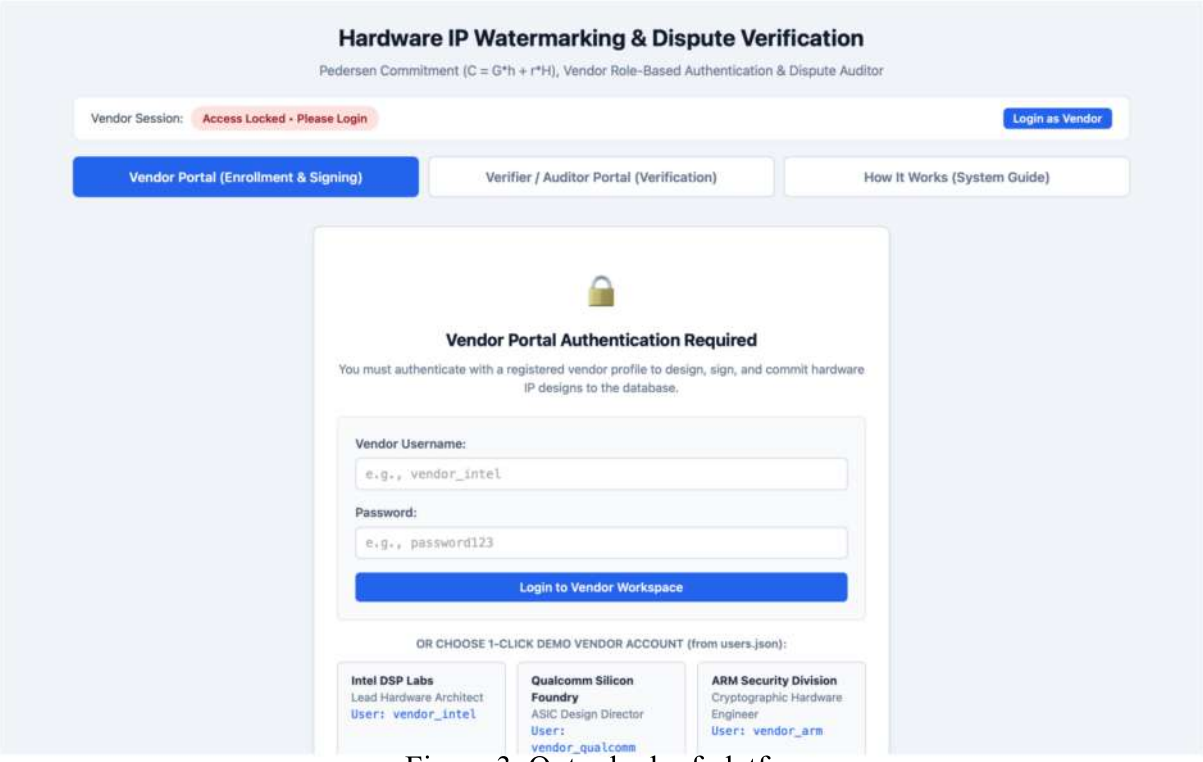


Figure 3. Outer look of platform

RESULT DISCUSSION AND ANALYSIS

Result: We have evaluated the theory of our project across the hardware impact, speed and other parameter. All this are discussed below:

Hardware Impact:

- **Circuit Overhead:** The designs need no extra registers i.e, zero overhead on hardware but in some scenario we have found that may be we need one or two more registers for embedding all the secret bits inside the chip IP.
- **Speed and accuracy :** Circuit overhead increases as negligible so the speed of the chip remains unaffected and accuracy is also unchanged.

System Speed (Execution Time):

We have measured the time taken for the cryptographic operation for our web platform created by python file. The verification phases takes less than 1 millisecond therefore our web site gives an instant verdict without any delay.

Dispute testing and Security:

We have tested the system across the different real world scenario to see if it correctly detects the stolen chips.

- Stolen Chip: The suspected chip have exact the same watermark and netlist hash and if the both public key and the pedersen commitment passes then the system confirms pirated and infringement of the design.
- Modified/Tampered Chip: If an attacker changes parts of the circuit, then the design hash changes and it causes the failure in digital signature.
- Fake claim: If someone tries to guess the watermark or the key then the mathematically check fails.

Web working:

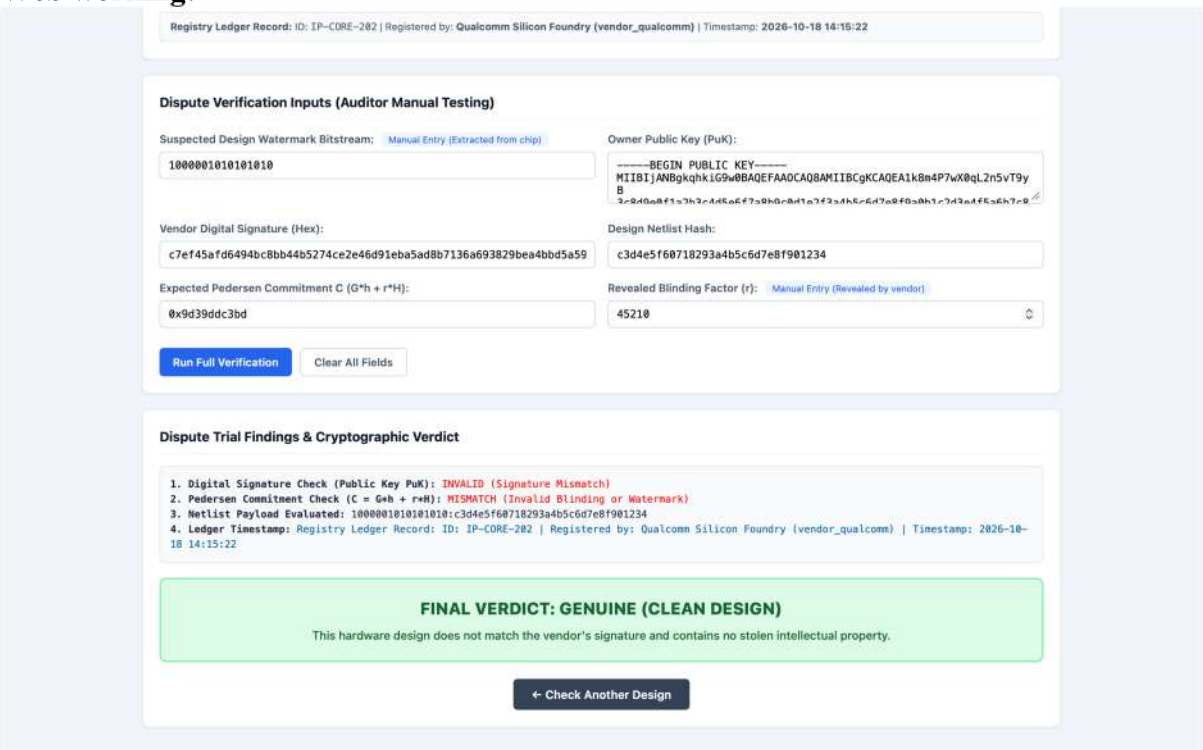


Figure 4: Verification of Clean design

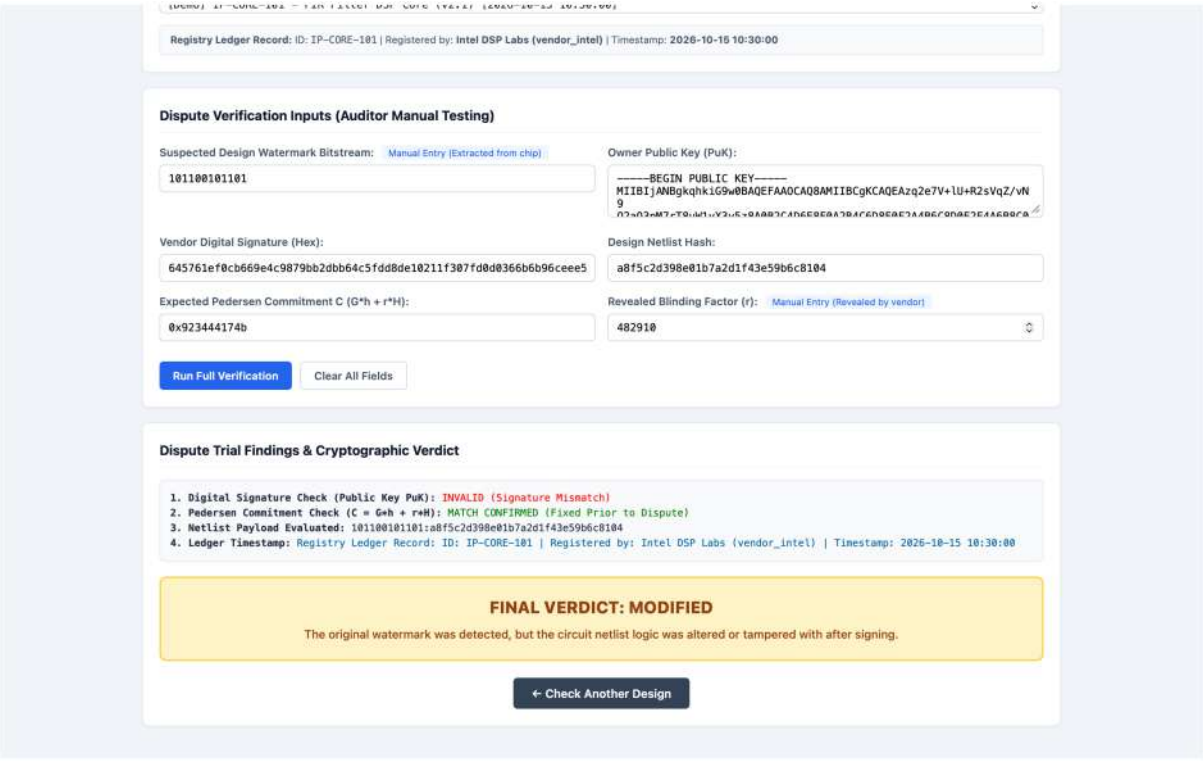


Figure 5. Verification of Modified design

Key Benefits:

- The Vendors Private key is never shared so it helps the vendor to keep it secret to themselves for lifetime.
- The pre registered Pedesen commitment proves that the vendor has already locked the watermark before the dispute happened.
- The web dashboard allows an auditor to verify multiple design back to back using simple clicks, i.e, web platform is very easy to use.

CONCLUSION AND FUTURE SCOPE

This project develops a security system to protect hardware designs using biometric data from the human biometric. The biometric patterns are used to create unique register allocation rules that act like a hidden watermark in the hardware design without adding extra hardware area, delay or power consumption.
It also uses cryptography and digital signatures to verify that a hardware design is genuine without revealing the owner's private biometric data or secret design rules. The web platform makes it easier to manage hardware ownership, verify authenticity and can detect fake or copied hardware designs.
The system combines asymmetric public key cryptography with Pedersen comitment to verify genuine designs. Through our interactive web development , the vendors can register custom IP into a local registry and on other side the independent auditors can verify the netlist, authenticity using only public key. The vendors private key is kept secret all the time because it has never been uploaded any where in the project.

Future scope:

The current system provides secure verification while keeping sensitive information private. In future it can be improved in the following ways:
1. Zero-knowledge Proofs (zk-SNARKS): Using tools like SnarkJS, the system can prove that the hidden rules in a hardware design are correct without revealing any secret variables or design details.
2. Decentralized Notarization: The strong verification hashes and timestamps on public blockchain will make the records more transparent and will reduce the dependence on a single central server.
3. Post Quantum Cryptography: Replacing digital signatures (RSA-2048) with quantum resistant algorithms such as ML-DSA (Dillithium) will help protect the system against future attacts from powerful quantum computers.
4. Automated RTL Netlist Parsing: On integrating an in browser netlist parser can directly read VHDL and also extract the register allocation table automatically. This will remove the need for manual bitstream and the hash input.

REFERENCES

[1] A.Kumar, V.M. Shrimal and R.Rubbina, “Analysing security and efficiency in a qunatum resistant decentralised E-voting system using blockchain, homomorphic encryption and zero knowledge proofs,” in Proc. 2nd Int. Conf. Res. Methodologies Knowl. Manage., Artif. Intell. Telecommun. Eng. (RMKMATE), Chennai, India,2025,pp.1-5,doi: 10.1109/RMKMATE64874.2025.11042466.

[2] D.Naidu, B.Wanjari, R.Bhojwani, S.Suchak,R.Baser, and N.K.Ray, “Efficient smart contract for privarcy preserving authentication in blockchain using zero knowledge proof,” in Proc. OITS Int. Conf. Inf. Technol. (OCIT), Raipur, India, 2023, pp. 969-974, doi: 10.1109/OCT59427.2023.10430710.

[3] S.Mishra and R.K. Dwivedi, “Securing records in healthcare IoT using zero knowledge proof based blockchain,” in proc. 3rd Int. Conf. IoT,Commun. Authom. Technol.(ICICAT), Gorakhpur, India,2025, pp. 1-5,doi: 10.1109/ICICAT68430.2025.11414459.

[4] P. Baptista, B. Pacheco, F. Apolinario, J.Silveria, N.Escravana, and C. Riberiro, “Zero knowledge programming cerifications,” in Proc. IEEE Int. Conf. Distrib. Ledger Technol. (ICDLT), Pune, India, 2025, pp. 1-5, doi: 10.1109/ICDLT66400.2025.11466848.

[5] N. Datta, “Zero knowledge one time digital signature scheme,” in Proc. IEEE Int. Conf. Comput. Intell. Comput. Res., Coimbatore, India, 2012, pp. 1-4, doi:10.1109/ICCIC.2012.6510192.

[6] A. Abraham, M. Hiremath, and F.H. Pohrmen, “Privacy preserving authentication using zero knowledge proofs,” in Proc. IEEE 17th Int. Conf. Comput. Intell. Commun. Netw. (CICN), Goa, India, 2025, pp. 1003-1008, doi: 10.1109/CICN67655.2025.11368139.

[7] R. Akash, B. Bafna, A. Raghavan, R.T. Jose, P.B. Honnavalli, and I.Radhakrishnan, “LoaclVault: Zero-knowledge fingerprint authentication without biometric storage,” in Proc. 8th Int. Conf. Comput. Commun. Internet (ICCCI), Okayama, Japan, 2026,pp, 238-243, doi:10.1109/ICCCI70321.2026.11666520.

[8] V. Bobade, S. Ladwan, P. Waghmare, L.Ghai, P. Chaudhari, and C.puri, “A paradigm shift in digital trust: A cybersecurity analysis of zero- knowledge proofs,” in Proc. Int. Conf. Modern Sustain. Syst. (CMSS), Shah Alam, Malaysia, 2026, pp. 1573-1576, doi: 10.1109/CMSS69636.2026.11689222.

[9] P. Singhal and N. Joshi, “Zero-knowledge proofs for privacy- preserving agricultural yield verification: A blockchain-based incentive system for sustainable farming methods,” in Proc. 17th Int. Conf. Contemp. Comput. (IC3), Noida, India, 2025, pp. 1-6, doi:10.1109/IC366947.2025.11290217.

[10] X.Wang, “Research on the recovery strategy of federated learning model base on zero-knowledge proof,” in Proc. Int. Conf. Algorithm, Artif. Intell. Comput. Vis. (AAICV), London, U.K., 2025, pp. 194-198, doi: 10.1109/AAICV66571.2025.00042.

[11] S.A. Butt et al., “if-ZKP: Intel FPGA-based acceleration of zero knowledge proofs,” in Proc. IEEE 32nd Annu. INt .Symp. Field-Programmable Custom Comput. Mach. (FCCM), Orlando, FL, USA,2024, p.212, doi: 10.1109/FCCM60383.2024.00033.

[12] J. Kuriakose, P.S Sisodia, Amruth V, D.K. Shah, and S.More, “Comparative study of diverse zero-knowledge argument systems,” in Proc. Int. Conf. Data Mining adv. Comput. (SAPIENCE), Ernakulam, India,2016,pp. 284-293, odi: 10.1109//SAPIENCE.2016.7684134.

[13] V.K. Polisetti, A. Bhattacharjee, A. Gupta, and S.Vollala, “zkSHIELD- A web based cross chain zero knowledge asset proof,” in Proc.15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT), Kamand, India,2024, pp. 1-5, doi: 10.1109/ICCCNT61001.2024.10726106.

[14] M. Chen, X. Qu, J. Cao, Z. Yang, and G.Sun, “Digital identity management based on blockchain and zero knowledge proof under the PIPL,” in Proc. 29th Int. Conf. Comput. Supported Cooperative Work Design (CSCWD),Fuzhou, China, 2026, pp. 340-345, doi: 10.1109/CSCWD68734.2026.11582607.

[15] V. Quach, R.Dantu, S. Talapuru, S.Zaman, and A. Pokharel, “ZCube: A zero-trust, zero-knowledge, and zero-memory platform for privacy and yet secured access,” in Proc. IEEE 6th Int. Conf. Trust, Privacy Secur. Intell. Syst. Appl. (TPS_ISA), Washington, DC, USA, 2024, pp. 166-175, doi: 10.1109/TPS-ISA62245.2024.00028.

[16] P.L. S.Kumari and A. Damodaram, “An alternative methodology for authentication and confidentality based on zero knowledge protocols using Diffe-Hellman key exchange,” in Proc. Int. Conf. Inf. Technol., Bhubneswar, India,2014, pp. 368-373, doi:10.1109/ICIT.2014.39.

[17] A. Pathak, T.Patil, S.pawar, P.Raut, and S. Khairnar, “Secure authentication using zero knowledge proof,” in Proc. Asian Conf. Innov. Technol. (ASIANCON), Pune, India, 2021, pp. 1-8, doi: 10.1109/ASIANCON51346.2021.9544807.

[18] M. Harikrishnan and K.V. Lakshmy, “Secure digital service payment using zero knowledge proof in distributed network,” in Proc. 5th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS). Coimbatore, India, 2019, pp. 307-312, doi:10.1109/ICACCS.2019.8728462.

[19] M. Hao, C. Shang, S.Wang, W.Jiang, and J.Nie, “UAV- assisted zero knowledge model proof for generative AI: A multiagent deep reinforcement learning approach,” IEEE Internet Things J., vol.. 12, no. 10. pp. 13441-13454, May 2025, doi: 10.1109/JIOT.2025.3531914.
[20] P. Jagwani and S. Kaushik, “Defending location privacy using zero knowledge proof concept in location based services,” in Proc. IEEE 13th Int. Conf. Mobile Data Manage., Bengaluru, India, 2012, pp. 368-371, doi: 10.1109/MDM.2012.23.
[21] G.L.de Oliveira, F. A. Mendonca, and R.V. Ramos, “Zero knowledge protocol in a radio over fiber network,” in Proc. SBMO/IEEE MTT-S Int. Microw. Optoelectron. Conf.. (IMOC), Belem, Brazil, 2009, pp. 834-837, doi:10.1109/IMOC.2009.5427462.
[22] S. Liu, “Privacy protection revolution: Zero-knowledge proof,” in Proc. Int. Conf. Data Anal., Comput. Artif. Intell. (ICDACAI), Zakopane, Poland, 2022, pp.394-397, doi: 10.1109/ICDACAI57211.2022.00084
[23] A.M. Allam, I.I. Ibrahim, A.Ali, and A.E.H. Elsawy, “Effiecient zero-knowledge identification scheme with secret key exchange ,” in Proc. 46th Midwest Symp. Circuits Syst., Cairo, Egypt, 2003, vol. 1, pp. 516-519, doi: 10.1109/MWSCAS.2023.1572331.
[24] A. Sengupta and Rahul Chaurasia, “Retinal biometric for securing JPEG-Codec hardware IP core for CE systems”, IEEE Transactions on Computer Electronics, vol.69, pp. 441-457, Aug. 2023, doi: 10.1109/TCE.2023.3264669.

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